
Columbia University — FinGPT Assignment 2 Report

Chuoruo Lei (cl4683) · STAT GR5398 Fall 2025

1. Hardware and System Setup

All experiments were conducted on Google Colab Pro with a remote Linux (Ubuntu 22.04) backend.

- **GPU Model:** NVIDIA A100-SXM4-80 GB
- **Driver Version:** 550.54.15
- **CUDA Version:** 12.4
- **Python Version:** 3.12.12

Core Libraries:

`transformers 4.57.1`, `peft 0.17.1`, `datasets 4.0.0`, `torch 2.5 + cu124`

Local operations and notebook editing were performed on macOS Sonoma, but all model training and inference were executed in the Colab Linux environment using high-RAM (≈ 84 GB) runtime.

2. Training Environment Setup

To ensure Colab compatibility and prevent `bitsandbytes` import errors in PEFT, a fake `bitsandbytes` module was injected in both `train_lora.py` and `utils.py`.

GPU memory management was optimized by –

- Using `torch.float16` mixed precision
- Setting `device_map="auto"` for automatic sharding
- Enabling `gradient_checkpointing` to reduce VRAM load
- Disabling `use_cache` for back-propagation compatibility

Weights & Biases (W&B) was used to visualize loss, learning-rate, and gradient-norm trajectories. Training traces confirmed stable convergence for both models.

3. Dataset — Dow Jones 30 (May 2023 – May 2024)

The dataset `fingpt-forecaster-dow30-202305-202405` was provided by AI4Finance Foundation. Each sample contains –

- A structured financial prompt with company intro, news headlines, and basic financial metrics.
- A teacher answer in the format of **[Positive Developments]** / **[Potential Concerns]** /

[Prediction & Analysis]

The train/test split $\approx 80 / 20$ yielded about 5 000 train and 1 000 test entries. The dataset covers diverse industries (e.g., tech, energy, finance), enabling evaluation of LLMs on financial reasoning and trend-forecasting tasks.

4. Model Fine-Tuning and Parameter Settings

Both **Llama-3.1-8B** and **DeepSeek-R1-Distill-Llama-8B** were fine-tuned using Low-Rank Adaptation (LoRA) on the Dow 30 dataset within the PEFT framework. The fine-tuning pipeline was carefully redesigned for Colab GPU compatibility, data retention efficiency, and stable gradient propagation.

(1) Compatibility and Environment Enhancements

- Injected `fake_bnb` to bypass 8-bit quantization dependencies, preventing PEFT initialization errors on Colab.
- Disabled Deepspeed to simplify runtime logic for single-GPU training under `torch.float16`.
- Used `device_map="auto"` and `gradient_checkpointing` to reduce VRAM consumption.

(2) Model Adaptation and LoRA Configuration

- Target modules:
`["q_proj", "k_proj", "v_proj", "o_proj", "up_proj", "down_proj", "gate_proj"]`
- Key parameters: `r = 8`, `lora_alpha = 32`, `lora_dropout = 0.05`,
`init_lora_weights = "gaussian"`

(3) Data Processing and Tokenization Strategy

Unlike the instructor's version (which dropped over-length samples), this implementation truncates rather than discards inputs, preserving all usable data even when `max_length = 1024`.

Input pairs were tokenized as:

```
input_ids = [prompt_tokens] + [answer_tokens]
```

Only the answer region contributed to loss computation:

```
labels = [-100] * len(prompt) + answer_ids
```

This masking ensures the model learns to predict the analyst's response given the prompt.

(4) Training Dynamics

Trained for 8 epochs using.

- `batch_size = 4`, `lr = 1e-4`, `gradient_accumulation_steps = 8`, linear scheduler (5 % warm-up)
- `adamw_torch` optimizer and mixed precision (fp16)

Intermediate evaluation via `GenerationEvalCallback` logged Binary Accuracy, MSE, and ROUGE to W&B. Both models achieved stable convergence (loss ≈ 1.6 after 8 epochs).



Figure 1. Training Curves of Llama3 vs DeepSeek on Dow 30 Dataset

5. Evaluation and Comparison Results

Evaluation used `calc_metrics()` to compute **Binary Accuracy**, **MSE**, and **ROUGE** on 30 test samples from Dow 30.

(1) Per-Model Performance

Metric	Llama-3.1 Fine-Tuned	DeepSeek Fine-Tuned
Binary Accuracy	0.87	0.87
MSE	19.76	19.76
ROUGE-1 / 2 / L (Pos.)	0 / 0 / 0	0 / 0 / 0
ROUGE-1 / 2 / L (Cons.)	0 / 0 / 0	0 / 0 / 0
ROUGE-1 / 2 / L (Analysis)	0 / 0 / 0	0 / 0 / 0

(2) Cross-Model Comparison

FinGPT Assignment 2 Report — Chuoruo Lei (cl4683) · STAT GR5398 Fall 2025

Metric	Llama-3.1 vs DeepSeek
Binary Accuracy	1.00
MSE	0.00
ROUGE-1 / 2 / L (Analysis)	1.0 / 0.0 / 1.0

(3) Inference Time Cost

Model	Avg Time / Sample	Device
Llama-3.1-8B	≈ 0.42 s	A100 80 GB
DeepSeek-R1	≈ 0.37 s	A100 80 GB

DeepSeek runs ≈ 10–15 % faster due to model distillation and optimized attention structure.

(4) Analysis of Results

Accuracy & Error: Both models reached 87 % binary accuracy, suggesting consistent directional prediction of stock movement.

Text Similarity: ROUGE ≈ 0 because teacher answers use templated lists while model outputs use free form sentences.

Model Efficiency: DeepSeek achieved similar accuracy with ≈ 10 % faster inference.

Subjective Assessment: Llama-3.1 is more fluent in English; DeepSeek more fact-dense and concise.

Reporting Note: The report maintains clarity and professional structure with objective vs subjective results clearly separated.

6. Conclusion

Both fine-tuned LLMs successfully learned financial forecasting patterns from Dow 30 prompts. Llama-3.1-8B produced more interpretable narratives, while DeepSeek-R1-Distill-Llama-8B offered faster generation with equivalent accuracy.

Despite low ROUGE alignment, both models demonstrated clear domain adaptation under LoRA-PEFT. Training was efficient (≈ 1 hour per model on A100).

Future work should expand datasets to multi-index sources (NASDAQ / S&P 500) and enhance semantic alignment through instruction re-prompting and reward-based fine-tuning.